



Tecnológico de Monterrey
Partial Exam 1 A
2026-Jan-May

Grade

Course: Calculus I

ID-number: _____

Professor: Fco. Javier Vazquez Tavares

Date: 18/02/2026

Last Names and Names: _____

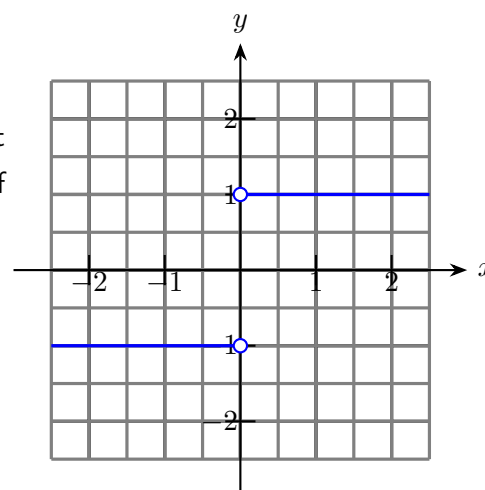
Group: 201

"In accordance with Tec de Monterrey Student Cod of Honor, in this exam I agree to conduct myself guided by academic honesty and not to use unauthorized or illegal means to do so."

Signature: _____

I Limits. Properties, Numerical, Graphic and Algebraic method.

1. (10 points) Applying the graphical technique to compute the limit when $x \rightarrow 0$. Argument if the limit converges. Use the notion of limit from the left and right.



2. (7 points) Compute the limit of the function $f(x) = \frac{6x^2+4}{5x^2-2}$ as $x \rightarrow \infty$ using the algebraic method.

3. (7 points) Compute the limit of the function $f(x) = \frac{\sqrt{x+9}-3}{x}$ as $x \rightarrow 0$ using the numerical method.

II Derivatives by definition.

1. (20 points) Compute the derivative of the function $f(x) = \frac{x^2-5x+6}{x-2}$ using the definition of derivatives at $x = 3$.

III Derivatives by rules.

1. (10 points) Using the power rule, compute the derivative of the function $f(x) = \frac{2}{3x^{3/2}}$.

2. (10 points) Using the product rule, compute the derivative of the function $g(x) = x2^x$.

3. (10 points) Using the quotient rule, compute the derivative of the function $g(x) = \frac{2^x}{x}$.

IV Derivative of trascendental functions.

1. (6 points) Applying the rules of derivatives for trascendental functions, compute the derivative of the function $f(x) = \ln(4x) - \cos(x)$.

V Multiple option section. Read the following questions and identify the option that best answers each, then write its letter on the line.

1. (5 points) _____ Select the option that is the definition of the derivative:
(a) $\lim_{h \rightarrow 0} \frac{f(x-h) - f(x)}{h}$ (b) $\lim_{h \rightarrow 0} \frac{f(x-h) + f(x)}{h}$ (c) $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
2. (5 points) _____ Can the quotient rule be considered a particular case of the product rule?
(a) No (b) It can be. Depends on the expression. (c) Not always. (d) Yes
3. (5 points) _____ Select the option with the range of the function $f(x) := x^2$
(a) $f(x) \in (-\infty, \infty)$ (b) $f(x) \in [0, \infty)$ (c) $f(x) \in (-\infty, 0]$ (d) $f(x) \in [-1, 1]$
4. (5 points) _____ Select the option with the numeric value of the operation of functions, $f(x) + g(x)$, with $f(x) := x^2 + x$ and $g(x) := x + 1$ at $x = 2$,
(a) 18 (b) 15 (c) 12 (d) 9
5. (5 points) _____ Select the alternative mathematical notation to express $f[g(x)]$
(a) $f \circ g$ (b) $g \circ f$ (c) $f \rightarrow g$ (d) $f(g)$
6. (5 points) _____ Select the option with the algebraic expression that shifts the linear function $f(x) := 2x$, 5 units downward,
(a) $f(x) := 2(x + 5)$ (b) $f(x) := 2(x - 5)$ (c) $f(x) := 2x + 5$ (d) $f(x) := 2x - 5$